Sri Chaitanya IIT Academy.,India.
A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI
A right Choice for the Real Aspirant
ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60(N&S), Elite, IAS & LIIT | JEE-MAIN-Shift-2 | Date: 27-03-2026
Time: 09.00Am to 12.00Pm | GTM-30 | Max. Marks: 300

KEY SHEET

MATHEMATICS

1	**4**	2	**1**	3	**1**	4	**3**	5	**3**
6	**3**	7	**1**	8	**3**	9	**2**	10	**4**
11	**1**	12	**1**	13	**3**	14	**3**	15	**2**
16	**2**	17	**4**	18	**2**	19	**2**	20	**4**
21	**16**	22	**3**	23	**30**	24	**93**	25	**252**

PHYSICS

26	**3**	27	**3**	28	**2**	29	**3**	30	**2**
31	**1**	32	**1**	33	**4**	34	**2**	35	**1**
36	**1**	37	**2**	38	**1**	39	**1**	40	**3**
41	**2**	42	**3**	43	**1**	44	**1**	45	**3**
46	**4**	47	**30**	48	**25**	49	**5**	50	**4**

CHEMISTRY

51	**1**	52	**3**	53	**2**	54	**2**	55	**4**
56	**3**	57	**3**	58	**4**	59	**3**	60	**1**
61	**2**	62	**4**	63	**1**	64	**3**	65	**3**
66	**2**	67	**3**	68	**1**	69	**4**	70	**1**
71	**400**	72	**1**	73	**10**	74	**4**	75	**203**

For More Join: @IITJEE_Advanced

SOLUTION
MATHEMATICS

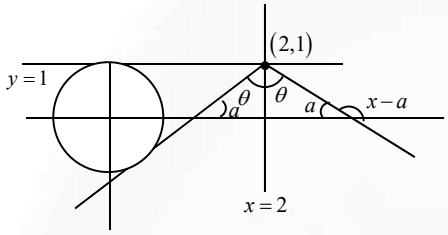
1. Equation of the ray: y – 1 = m(x-2)

mx –y -2m + 1 =0

$\therefore$ touches the circle $\therefore\ p = r \Rightarrow \left|\frac{1-2m}{\sqrt{1+m^2}}\right| = 1$

$\Rightarrow 1+4m^2-4m = 1+m^2 \qquad \because m \neq 0 \Rightarrow 4m-4 = m$

$\Rightarrow m = \frac{4}{3} \quad \tan\alpha\frac{4}{3} \Rightarrow \tan(\pi-\alpha) = -\frac{4}{3}$



$\therefore$ Equation of incident ray $y-1 = -\frac{4}{3}(x-2)$

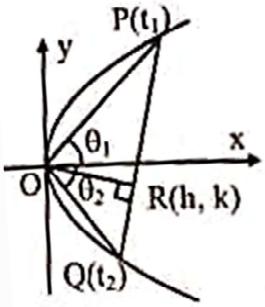
$\Rightarrow 3y-3 = -4x+8 \Rightarrow 4x+3y = 11$ ans.

2. Chord with feet of the perpendicular as (h,k) is $hx+ky = h^2+k^2$

Homogenise $y^2 = 4ax$ with the help of (1) and use coefficient of

x^2 + coefficient of $y^2 = 0 \quad \tan\theta_1 = \frac{2}{t_1}; \tan\theta_2 = \frac{2}{t_2}$

$\frac{2}{t_1}\frac{2}{t_2} = -1, \quad \therefore t_1t_2 = -4$



$\therefore$ Equation of chord $PQ\ 2x-(t_1+t_2)y-8a = 0$

Slope of R ×slope of PQ =-1 $\quad \therefore \frac{k}{h}\left(\frac{2}{t_1+t_2}\right) = -1 \qquad \therefore t_1+t_2 = \frac{-2k}{h}$

$\therefore$ equation of chord PQ will be $2x-\left(\frac{-2k}{h}\right)y-8a = 0 \qquad hx+ky = 4ah$

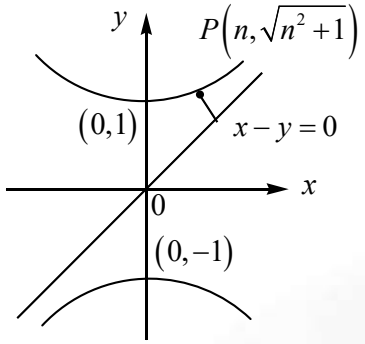
(h,k) line of this line)

$\therefore h^2+k^2 = 4ah$

$\therefore$ locus of R (h,k) is x^2+y^2-4ax

3. Curves is rectangular hyperbola.

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Perpendicular distance $d_n=\left|\frac{n-\sqrt{n^2+1}}{\sqrt{2}}\right|$

$\lim_{n\to\infty}(n\cdot d_n)=\lim_{n\to\infty}\frac{n}{\sqrt{2}}\left|\left(\sqrt{n^2+1}-n\right)\right|$ $\lim_{n\to\infty}\frac{n}{\sqrt{2}}\left|\frac{1}{\sqrt{2^2+1}+n}\right|=\frac{1}{2\sqrt{2}}$.

4. We have S = ABCD=A(BCD)= AA^T .....(1)
Now S^3 = SSS = (ABCD) (ABCD) (ABCD)
= (ABC) (DAB) (CDA) (BCD) = $D^TC^TA^T=(BCD)^TA^T=\left(A^T\right)^TA^T$
$=AA^T$ .....(2)
$\therefore$ From (1) & (2) we get
$S=S^3\Rightarrow S-1$ is correct
Now, on multiplying b.t.s. by S,
We get $S^2=S^4\Rightarrow$ S-2 is correct.

5. (A) given $P(A\cap B)=\frac{1}{12}\Rightarrow P(A),P(B)=\frac{1}{12}$.....(1)
And $P\left(\overline{A}\cap B\right)=P\left(\overline{A}\right)P\left(\overline{B}\right)=\frac{1}{2}\quad\Rightarrow\left(1-P(A)\right)\left(1-P(B)\right)=\frac{1}{2}$
Solving equation (1) and equation (2) and using P(A) < P(B)
We have $P(A)=\frac{1}{4}P(B)=\frac{1}{3}\qquad\therefore P(A-B)=P(A)-P(A\cap B)$
$=\frac{1}{4}-\frac{1}{4}\times\frac{1}{3}=\frac{2}{12}=\frac{1}{6}\Rightarrow$ (A) is correct.
Method –I
After three heads tail can occur is fourth toss with probability = $\frac{1}{2}\Rightarrow$(B) is correct.
Method –II
Required probability = $P\left(\frac{A}{B}\right)=\frac{P(A\cap B)}{P(B)}$
Where B is event of occurrence of first 3 heads and B is event of occurrence of tail on fourth trail.
$\therefore P(A/B)=\frac{\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}}{\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}}=\frac{1}{2}\Rightarrow$(B) is correct.
C) numbers of ways I which two E's do not occur adjacently
= total number of ways – when two E's occur

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Simultaneously $=\frac{71}{21}-61=\left(\frac{7}{2}-1\right)61=\frac{5}{2}\times 61$

$\therefore$ Required probability $=\frac{\frac{5}{2}\times 6!}{\frac{7!}{2!}}=\frac{5}{2}\times\frac{6!\times 2}{7.6!}=\frac{5}{7}$ $\Rightarrow$(C) is incorrect.

(D) As $0\le P(A\cup B)\le 1$

$0\le P(A)+P(B)-P(A\cap B)\le 1$

Hence $P(A)+P(B)-1\le P(A\cap B)\le P(A)+P(B)=$

(D) is correct.

6. $\frac{d}{dx}\left(\frac{g(x)}{g(g(x))}\right)at\ x=4$

$=\frac{g(g(x))g'(x)\cdot -g(x)g'(g(x))\cdot g'(x)}{(g(g(x)))^2}\Big]_{x=4}$

Now, $f(0)=4\Rightarrow f^{-1}(4)=g(4)=0$

And $g'(4)=\frac{1}{f'(0)}=\frac{1}{3}$

Also, $f(-1)=0\ \Rightarrow f^{-1}(0)=g(0)=-1$

$=\frac{g(g(4))g'(4)-g(4)g'(g(4))g'(4)}{(g(g(4)))^2}=\frac{-1}{3}$.

7. $f(2)<0$ $4+4a+3a+10<0$

$7a<-2$ $a(-(-\infty -2)$.

8. A) No. of such triangles = 10 ${}^6C_1$ + = 60

B) No. of such triangles = 10

C) No. Of such quadrilaterals = for n > 5 $\frac{3n(n-5)}{2}$

D) No. of such quadrilaterals = 10 (when four consecutive points are taken)

9. $A(t)=\int_{-2}^{t} f(x)dx$ $\Rightarrow A'(t)=f(t)$

$\therefore A'(3)=f(3)$ $\therefore \lim_{t\to 3^-}\frac{A(3)-A(t)}{3-t}=A'(t)=f(3)$ $\Rightarrow f(3)=100$.

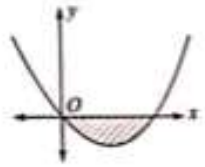
10. $\frac{dy}{dx}=\frac{(x+1)^2+y-3}{x+1}$ $\therefore \frac{dy}{dx}=\frac{1}{x+1}y+\frac{x^2-2x-2}{x+1}$

$I.F=e^{-\int\frac{dx}{1+x}}=e^{\ln(1+x)}=\frac{1}{1+x}$ $\therefore y\frac{1}{1+x}=\int\frac{x^2+2x-2}{x+1}\frac{1}{x+1}dx\int\frac{(x+1)^2-3}{(x+1)^2}dx$

$=\int\left(1-\frac{3}{(x+1)^2}\right)dx$ $\frac{y}{1+y}=x+\frac{3}{x+1}+C$

When $x=2,y=0,0=2+1+c\Rightarrow c=3$

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$\therefore \frac{y}{x+1} = x + \frac{3}{x+1} - 3 = x + \frac{3-3x-3}{x+1} = \frac{x(x+1)-3x}{x+1}$

$y = x^2 + x - 3x = x^2 - 2x \qquad y = x^2 - 2x$

$So,\ A = \left|\int_0^2 (x^2 - 2x)\,dx\right| = \frac{4}{3}$

11. $2x \le 3y + 1$

$$R\begin{Bmatrix}(2\ 2)(2\ 3)(2\ 5)(2\ 7)(2\ 9)\\(3\ 2)(3\ 3)(3\ 5)(3\ 7)(3\ 9)\\(5\ 3)(5\ 5)(5\ 7)(5\ 9)\\(7\ 5)(7\ 7)(7\ 9)\\(9\ 7)(9\ 9)\end{Bmatrix}$$

$\lambda = 1$

To make R symmetric add (5 2) (7 2) (9 2) (7 3) (9 3) (9 5)

M= 6

12. $\left|\frac{z_1 2z_2}{2 - z_1 z_2}\right| = |1 \Rightarrow 2z_2|^2 = |2 - z_1\bar{z}_2|^2 \Rightarrow (z_1 - 2z_2)(\bar{z}_1 - 2\bar{z}_2) = (2 - z_1\bar{z}_2)(2 - \bar{z}_1 z_2)$

$\Rightarrow |z_1|^2 + 4|z_2|^2 - 4 - |z_1|^2|z_2|^2 = 0 \Rightarrow 4(|z_2|^2 - 1) - |z_1|^2(|z_2|^2 - 1) = 0$

$\Rightarrow |z_1|^2 - 4 = 0 \Rightarrow |z_1| = 2$ .is a circle of radius 2 and centre at origin.

13. We have $3x + y + \beta z = 3$

$2x + ay - z = -3 \qquad x + 2y + z = 4$

From (1) -3 (3) , we get $-5y + (\beta - 3)z = -9$

From (2) -3 (3) we get $(\alpha - 4)y - 3z = -11$

For infinite solution, we must have

$\frac{-5}{\alpha - 4} = \frac{\beta - 3}{-3} = \frac{-9}{-11} \quad \therefore \alpha = \frac{-19}{9}, \beta = \frac{6}{11} \quad \Rightarrow 22\beta - 9\alpha = 31.$

14. Function is continuous at x = 0 $\therefore L.H.L = R.H.L = 4(at\ x = 0)$

$L.H.L = 2\lim_{x\to 0} \frac{\sin(k_1+1)x + \sin(k_2-1)x}{x} = 2((k_1+1) + (k_2-1)) = 2(k_1 + k_2)$

$R.H.L = 2\lim_{x\to 0} \frac{2}{x} \log_e\left(\frac{2 + k_1 x}{2 + k_2 x} x\right) = 22\lim_{x\to 0} \frac{\log_e\left(1 + \frac{(k_1 - k_2)}{2 + k_2 x}\right)}{\frac{(k_1 - k_2)}{2 + k_2 x} x} - \frac{(k_1 - k_2)}{2 + k_2 x} = k_1 - k_2$

$\therefore 2(k_1 + k_2 = k_1 - k_2) = 4 \qquad \Rightarrow k_1 + k_2 = 2\ and\ k_1 - k_2 = 4$

$\therefore k_1 = 3, k_2 = -1 \qquad So,\ k_1^2 + k\frac{2}{2} = 9 + 1 = 10.$

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15. $\left|\left(\vec{b}\times\vec{a}\right)-\vec{b}\right|^2=\left|\left(\vec{b}\times\vec{a}\right)\right|^2+\left|\vec{b}\right|^2 \qquad \left(\because \vec{b}\perp\vec{b}\times\vec{a}\right)$

$=4+1=5.$

16. Let orthocenter, centroid and circumcenter of triangle are H, G and O respectively.

$\therefore$ position vector of G is…

$$\frac{\left(4\vec{p}+\vec{q}-3\vec{r}\right)+\left(-5\vec{p}+\vec{q}+2\vec{r}\right)+\left(2\vec{p}+2\vec{r}\right)}{3}$$

Or $\frac{\vec{p}+\vec{q}+\vec{r}}{3}$

Now G divides HO in the ratio 2 : 1.

$\therefore$ position vector of G is $\frac{1.\frac{\vec{p}+\vec{q}+\vec{r}}{4}+2\left(\alpha\vec{p}+\beta\vec{q}+\gamma\vec{r}\right)}{1+2}$

So, $\frac{\vec{p}+\vec{q}+\vec{r}}{12}+\frac{2\left(\alpha\vec{p}+\beta\vec{q}+\gamma\vec{r}\right)}{3}=\frac{\vec{p}+\vec{q}+\vec{r}}{3}$

$\therefore \alpha+\beta+\gamma=\frac{3}{8}$

$\alpha+2\beta+5\gamma=3.$

17. Equation of line L_2 *is* $\frac{x+4}{3}=\frac{y-4}{2}=\frac{z-3}{0}$

$\therefore$ Shortest distance between L_1 and L_2

$$=\frac{\begin{vmatrix}1-(-4) & -1-4 & -4-3\\ 2 & -3 & 2\\ 3 & 2 & 0\end{vmatrix}}{\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & -3 & 2\\ 3 & 2 & 0\end{vmatrix}}$$

$$=\frac{\left|5(0-4)+5(0-6)-7(4-(-9)\right|}{\left[-4\hat{i}+6\hat{j}+13\hat{k}\right]}$$

$$=\frac{141}{\sqrt{16+36+169}}=\frac{141}{\sqrt{122}}.$$

18. $f(x)=6+4\cos 3x\sin 3x\cos 6x=6+\sin 12x$

So, the range of $f(x)$ *is* $[5,7]$

Therefore, $(\alpha,\beta)=(5,7)$

Hence, the required distance is $\left|\frac{15-28-12}{5}\right|=5.$

19. $\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}}\left(\frac{1}{[x]+5}\right)dx$

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$$\int_{\frac{-\pi}{2}}^{-1}\frac{1}{3}dx+\int_{-1}^{0}\frac{1}{4}dx+\int_{0}^{1}\frac{1}{5}dx+\int_{1}^{\frac{\pi}{2}}\frac{1}{6}dx$$

$$\frac{1}{3}\left[-1+\frac{\pi}{2}\right]+\frac{1}{4}(1)+\frac{1}{5}(1)+\frac{1}{6}\left(\frac{\pi}{2}-1\right)$$

$$\left(\frac{\pi}{2}-1\right)\left(\frac{1}{3}+\frac{1}{6}\right)+\frac{9}{20}$$

$$\left(\frac{\pi}{2}-1\right)\frac{1}{2}+\frac{9}{20}$$

$$\frac{\pi-2}{4}+\frac{9}{20}= \quad \frac{5\pi-10+9}{20}=\frac{5\pi-1}{20}=\frac{\pi}{4}-\frac{1}{20}.$$

20. Let $\sin^{-1}x=\theta$ *or* $x=\sin\theta$

$\therefore \cos(2\theta)=\frac{1}{9}$

$\Rightarrow 1-2\sin^2\theta=\frac{1}{9}$

$\Rightarrow \sin=x=\frac{2}{3}$ (as m and n are co-prime natural numbers)

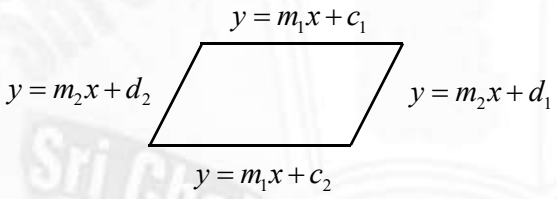
So, given quadratic equation becomes $2x^2-3x+1=0$

Roots of the equation are $\alpha=1,\beta=\frac{1}{2}$

Clearly, point $\left(1,\frac{1}{2}\right)$ lines on line $5x+8y=9$.

21. Area of the parallelogram $\frac{|c_1-c_2||d_1-d_2|}{|m_1-m_2|}$ (for the figure as shown)

Hence area of the parallelogram formed by the 1st set of 4lines $\frac{|c-d|^2}{|a-b|}=18$.....(1)



And area of the parallelogram formed by the 2nd set of 4 lines $\frac{|c+d|^2}{|a-b|}=72$.....(2)

From (1) $|c-d|^2=|a-b|\cdot 18$

From (2) $|c+d|^2=|a-b|\cdot 72$

Since a, b, c, $d\in N$

Hence RHS must be a prefect square for some least value of $|a-b|$ which obviously can be equal to 2 (think?) $\Rightarrow |a-b|=2$

And a = 1 and b = 3

$$\left.\begin{array}{l} Hence\ c-d=6 \\ and \quad c+d=12 \end{array}\right]$$

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C = 9 and d = 3

$\therefore (a+b+c+d)=16$.

22. We have $f(x)=\alpha e^{2x}+\beta e^{x}-\ell x$

$Now\ f(0)=-1\Rightarrow \alpha+\beta=1$.....(1)

$f'(\ell n 2)=30\Rightarrow 8\alpha+2\beta-\gamma=30$.....(2)

And $\Rightarrow \int_0^{\ell n 4}\left(f(x)+\gamma x\right)dx=24\Rightarrow \int_0^{\ell n 4}\left(\alpha e^{2x}+\beta e^{x}\right)dx=24$

$$\Rightarrow\left(\frac{\alpha e^{2x}}{2}+\beta e^{x}\right)_0^{\ell n 4}=24 \qquad \Rightarrow(8\alpha+4\beta)-\left(\frac{\alpha}{2}+\beta\right)=24$$

$\Rightarrow 15\alpha+6\beta=48$

On solving (1), (2), (3), we get

$\alpha=6,\beta=-7,\gamma=4$

Hence $(\alpha+\beta+\gamma)=6-7+4=3$..

23. $n+5_{cr-2}=5k$

$n+5_{cr-1}=10k$

$n+5_{cr}=141k$

$n-3r=9\ \ 5n-12r=-30$

24. $p_n=729,81,9$…….n term

$3^6\,3^4\,3^2.....3^{-2n+8}$

$P_n=3^{6+4.....(-2n+8)}=3^{n(7-n)}$

$(P_n)^{\frac{1}{n}}=3^{(7-n)}$

$=3^6+3^5.....50$ Terms

$$3^6\frac{\left[1-\left(\frac{1}{3}\right)^{50}\right]}{1-\frac{1}{3}}$$

$$3^6\frac{\left[3^{50}-1\right]\times 3}{3^{50}\times 2}=\frac{3^{50}-1}{3^{43}\times 2}$$

25. $f(x)=-\left(p^2-6p+8\right)\cos 4x+2(2-p)x+7$

$f'(x)=4\left(p^2-6p+8\right)\sin 4x+(4-2p)\neq 0$

$$\sin 4x\neq\frac{2(p-2)}{4(p-4)(p-2)}\quad p\neq 2$$

$p\in\left(\frac{7}{2}\ \frac{9}{2}\right)$.

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PHYSICS

26. Velocity transverse waves is $v=\sqrt{\frac{T}{\mu}}$ or $\upsilon^2\frac{T}{\mu}$

Where $\mu=\frac{3\times10^{-2}}{1.5}=2\times10^{-2}kg/m$

$T=(180)^2\times2\times10^{-2}=648N$.

27. As the fisherman is air, the rays of light travels from rarer to denser medium, they bends towards the normal. So, the fisherman appears taller

Yes, the apparent depth decreases, further, when water tank viewed obliquely as compared to the depth when see near normally.

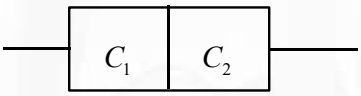
28. Wave length light $\lambda=589\times10^{-9}$ m

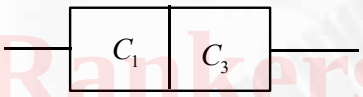
Frequency of reflected light

$v=\frac{c}{\lambda}=\frac{3\times10^8}{589\times10^{-9}}=5.09\times10^{+14}Hz$:

29. $Q=Q_1+Q_2=\frac{q}{8\in_0}+\frac{q}{\in_0}=\frac{9q}{8\in_0}$

30.

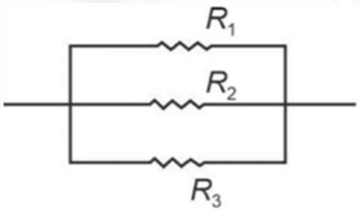




$$C=\frac{\varepsilon_0\frac{A}{2}}{\frac{d}{2k_1}+\frac{d}{2k_2}}+\frac{\varepsilon_0\frac{A}{2}}{\frac{d}{2k_1}+\frac{d}{2k_2}}=\frac{\varepsilon_0A}{d}\left\{\frac{1}{\frac{1}{2}+\frac{1}{3}}+\frac{1}{\frac{1}{2}+\frac{1}{5}}\right\}$$

$C=C_0\left(\frac{6}{5}+\frac{10}{7}\right)$ $C=C_0\left(\frac{92}{35}\right)=\frac{NC_0}{3}$ $N=\frac{276}{35}\approx8$

31.



$R_1=\frac{\lambda2\pi r}{4}=\frac{\lambda\pi r}{4}=\frac{\lambda\pi r}{2}$ $R_2=\lambda2R$

$R_3=\frac{3}{4}\lambda2\pi r=\frac{3\lambda\pi r}{2}$ $\frac{1}{R}=\frac{1}{\lambda\pi r}+\frac{1}{\lambda2r}+\frac{2}{3\lambda\pi r}$

$\frac{1}{R}=\frac{12+3\pi+4}{6\lambda\pi r}$ $R=\frac{6\pi\lambda r}{3\pi+16}$.

32. $e=Blv\sin\theta=0.6\times10^{-4}\times10\times5=3\times10^{-3}V$

According to the Fleming's right hand rule, the force west is down, then the direction of induced emf will be from west to east.

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33. a) False, The total momentum and total energy of the system are conserved and not of each body.
b) False, the total energy is constant if the internal forces doing work are conservative and external forces do no work.
c) False, work done over a closed loop is zero, for conservative forces only.
d) True, because in an in-elastic collision, some K.E usually changes into some other form of energy

34. $\frac{c}{v} = n\sqrt{\mu_r \varepsilon_r} \quad \Rightarrow \frac{c}{v} = \sqrt{6}$

35. $\frac{1}{\lambda} = R\left[\frac{1}{1} - \frac{1}{3^2}\right] = \frac{8R}{9}$

$\lambda_1 = \frac{9}{8R} = 5400A^0$

$\frac{1}{\lambda} = R\left[\frac{1}{1} - \frac{1}{2^2}\right] = \frac{3R}{4} \Rightarrow \lambda_2 = \frac{4}{3R}$

$\frac{\lambda_2}{\lambda_1} = \frac{4}{3R} \times \frac{8R}{9} \Rightarrow \lambda_2 = 6400A^0.$

36. Conceptual

37. $y = AA * \overline{AB} \cdot \overline{AB} \qquad = A * \overline{AB}$

$= A * \left(\overline{A} + \overline{B}\right) \qquad = A\overline{A} + A\overline{B}$

38. a) when we below over a piece of paper, the velocity of air above the paper increase. So, in accordance with Bernoulli's theorem $\left(P + \frac{1}{2}\rho v^2 = cons\tan t\right)$, the pressure of air decreases above the paper. Due to the pressure difference of air between below and above the paper a lifting force acts on paper and hence it remain horizontal.
b) A spinning ball experiences dynamic lift apart from gravity and hence it does not follow a parabolic path; it is rather curved. A ball follows a parabolic path only when it is subject to gravity along and that happens when the ball is thrown (horizontally without spin.

39. $F\left(\frac{ML^2T^{-1}\left(LT^{-1}\right)}{M^{-1}C^3T^{-2}L}\right)^{\frac{1}{2}} = MLT^{-2}$ = Force.

40. $V_{avg} = \frac{S_O ABC}{time} = \frac{5}{3} = m/s$

At t = 4, slope = 0 v = 0

$V_{avg} = \frac{s}{t} = \frac{10}{4} = 2.5$

$\Delta V = 0$ then a = 0

41. $a = \frac{g}{2} \quad t = \sqrt{\frac{2s}{a}} = \sqrt{\frac{2\times10\times10^{-2}}{10/2}} = 0.2\sec$

42. $I - \frac{4MR^2}{2} - \left(\frac{MR^2}{4\times2} + \frac{MR^2}{4}\right)$

$= \frac{16MR^2 - 3NR^2}{8} = \frac{13MR^2}{8}$

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43. Conceptual

44. $Pv^2 = RT = 1P = \frac{RT}{v^2}$ $w = \int_{v_1}^{v_2} pdv = \int_{v_1}^{v_2} \frac{RT}{v^2} dv$

$= -\left(\frac{1}{v} RT\right)_2^4 = -\left(\frac{RT}{4} - \frac{RT}{2}\right) = \frac{RT}{4}$

45. Speed of the light,

$c = \frac{E_0}{B_0} \Rightarrow E_0 = cB_0 \quad E_0 = \left(3\times10^8\right)\left(3\times10^{-8}\right)$

$E_0 = 9\ Vm^-$ As $\vec{B}$ = along y –axis

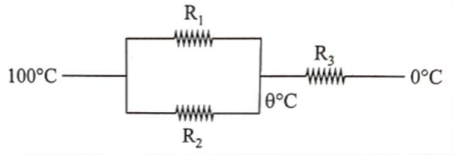
$\vec{v}$ = along negative x-axis

Hence $\vec{E}_0$ = along z -axis

46. $mg\cos\theta = \frac{mv^2}{R} \qquad g\cos\theta = \frac{2g(R-n)}{R}$

$h = \frac{2R}{3} = 1h = \frac{2\times6}{3} = 4.$

47.



$R_1 = \frac{2\ell}{k_1 A}, R_2 = \frac{2\ell}{k_2 A}, R_3 = \frac{\ell}{k_3 A}$ $(100-\theta)\left(\frac{1}{R_1} + \frac{1}{R_2}\right) = \frac{(\theta-0)}{R_3}$

$(100-\theta)\frac{(k_1+k_2)A}{2\ell} = \frac{\theta k_3 A}{\ell}$ $(100-\theta)(k_1+k_2) = 2\theta k_3$

$\theta\frac{100(k_1+k_2)}{k_1+k_2+2k_3} = \frac{100\times180}{450} = 40$

48. Least count of screw gauge

$= \frac{\text{Pitch}}{\text{Number of circular scale divisions}} \qquad \Rightarrow \text{LC} = \frac{0.75\text{mm}}{25} = 0.03\text{mm}$

Length l = 6 mm

Breadth b = 1.5 mm

Area = lb $\Rightarrow$ % error in area

(% error in length) + (% error in breadth)

$= \frac{\Delta\ell}{\ell}\times100 + \frac{\Delta b}{b}\times100$ $\frac{dA}{A}\% = \frac{0.03}{6}\times100 + \frac{0.03}{2.5}\times100 = 2.5\%.$

49. $B = N(\mu_0 ni)$

$N = \frac{B}{\mu_0 ni} = \frac{2.5\times w^{-2}\times80\times11^{-2}}{4\times\pi\times400\times8} \qquad = 4.97 \simeq 5$

50. $T = 2\pi\sqrt{\frac{m}{L^2 \in g}}$

$= 2\pi\sqrt{\frac{L_{10}\times10^{-3}}{100\times10^{-4}\times10^3\times10}} \qquad = 4\pi\times10^{-2}\ \text{sec}$

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CHEMISTRY

51. $\pi = CRT \qquad \pi = \frac{wt}{GMW} \times \frac{1000}{V(ml)} \times R \times T$

$GMW = \frac{wt}{\pi} \times \frac{1000}{V(ml)} \times R \times T$

$= \frac{1.26 \times 1000 \times 0.083 \times 300}{2.57 \times 10^{-3} \times 400} \qquad = 30,519.4 gm.$

52. The metallic radius of Na is 1.86 A^0 and the ionic radius of Na^+ is lesser then 1.86A^0 cations are small in size and anions are larger in size than their parent atoms.

53. XeO_3, sp^3 Pyramidal.

$[PtCl]^{-2}\ dsp^2$ square planar

$BrF_5\ , sp^3d^2$ square Pyramidal.

$SF_4 - sp^3d$ see saw

54. $FeO_{(g)} + CO_{(g)} \rightleftarrows Fe_{(s)} + CO_{2(g)}$

Initial $1.4\,atm \qquad 0.8\,atm$

$Q_P = \frac{P_{CO_2}}{P_{CO}} = \frac{0.8}{1.4} = 0.571.$

Since $Q_p > K_p$. This means that the reaction will move in the backward direction to attain the equilibrium therefore, partial pressure of CO_2 will decrease while that of CO will increase.

$P_{CO_2} = (0.8 - p)\,atm \quad P_{CO} = (1.4 + p)\,atm$

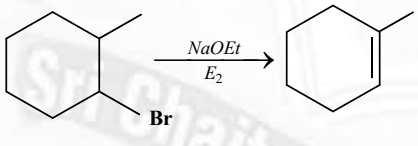
$K_p = \frac{P_{CO_2}}{P_{CO}} \qquad 0.465 = \frac{0.8 - p}{4.4 + p}$

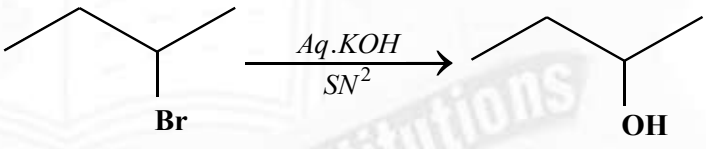
$0.651 + 0.465p = 0.8 - p$

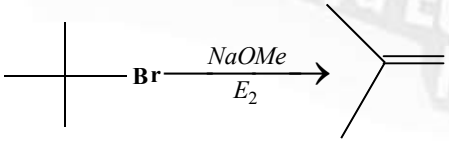
$1.465p = 0.8 - 0.651 \quad p = 0.101$

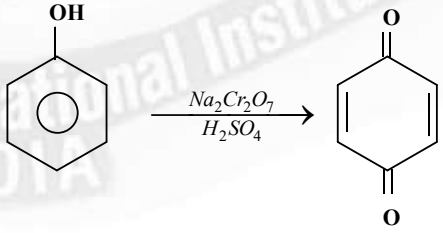
$P_{CO_2} = 0.8 - p \ = 0.8 - 0.101 \quad P_{CO_2} = 0.699.$

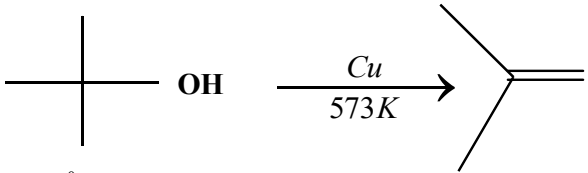
55.

A) 

B) 

C) 

D) 

E) 

(3^0 **Alcohol**)

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56. a) N_2 is very inert at room temp due to the strong triple bond between the nitrogen atoms
b) The common oxidation states of these elements -3, +3 and +5.
c) Nitrogen has unique ability to form Pπ - Pπ multiple bonds
d) The stability of +3 oxidation state increases down the group due to inert pair effect
e) Nitrogen can exhibit maximum covalency of 4.

57. Rate $= K[A]^2[B]^0 \quad = K[2A]^2[2B]^0 \quad = 4K[A]^2[B]^0$
Reaction rate 4 time increased x = 4
the first order rate equation is

$$K = \frac{2.303}{t}\log\frac{[R]_0}{[R]} \qquad \frac{Kt}{2.303} = \log\frac{[R]_0}{[R]}$$

$$\log\frac{[R]_0}{[R]} = \frac{K}{2.303}t \qquad Y = mX \quad y = 1 \quad x + y = 5.$$

58. Gabriel phthalimide is used to prepare RNH_2 as it involves S_{N^2} reaction which aryl halides do not undergo.

59. 1) The correct order is decided by hyper conjugation
2) cis alkene has higher B.P
3) Stability of alkene using hyper conjugation and extended conjugation
4) The correct order is decided by hyperconjugation.

60. Sodium dichromate is more soluble than potassium dichromate
V_2O_3 more basic
V_2O_4 less basic
V_2O_5 Amphoteric.

61. $4HCl_{(aq)} + MnO_{2(s)} \rightarrow 2H_2O_{(l)} + MnCl_{2(aq)} + Cl_{2(g)}$

$4\times 36.5gm \quad 87gms$
$? \quad 50gm$

Mass of $HCl = \frac{4\times 36.5\times 50}{87} = 83.9gms$

62. $[Co(NH_3)_6]^{+3} - d^2sp^3 \qquad [CrF_6]^{-3} - d^2sp^3$
$[CoF_6]^{-3} - sp^3d^2 \qquad [Fe(CN)_6]^{-3} - d^2sp^3 \quad SF_6 - sp^3d^2 \qquad PCl_6^- - sp^3d^2$

63. A is stabilized by hyper conjugation
B and C stabilized by resonance and hyper conjugation.

64. Amylose water soluble. Component which constitutes about 15-20 % starch amylopectin is insoluble in water and constitutes about 80-85 % of starch.

65. $r = 0.529\times\frac{n^2}{z}\ A^0 \qquad \Delta R_1 = \frac{0.529}{3}(4^2 - 3^2)$

$= \frac{0.529}{3}\times(16-9) \qquad = \frac{0.529}{3}\times 7\ A^0$

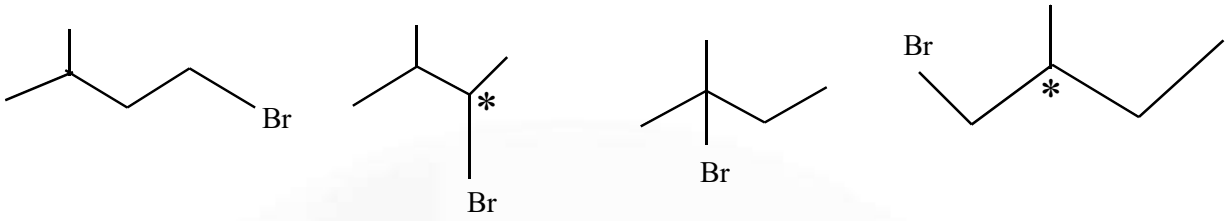
$\Delta R_2 = \frac{0.529}{2}(4^2 - 3^2) \qquad = \frac{0.529}{2}\times 7 \qquad \frac{\Delta R_1}{\Delta R_2} = \frac{2}{3}$

66. $E.N$ of group 13 does not decrease down the group
All elements do not show +3 stable oxidation state.

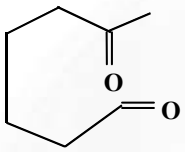
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67. $RBr \xrightarrow[\text{dryether}]{Mg} RMgBr \xrightarrow{H_2O} RH$

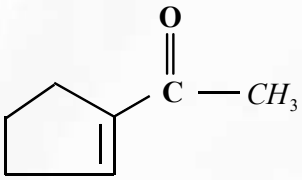
RBr can be

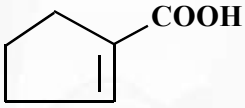


68.

A is 

(intra molecular aldol condensation)

B is 

C is 

69. i) *B.P* He < Ne < Ar < Kr < Xe

ii) electron gain enthalpy with –ve sign S > Se > Te > O

iii) Acid order Hl > HBr > HCl > HI

iv) oxides stability I > Cl > Br

v) ionic character MF > MCl > MBr > MI

70. $2N_2O_{5(g)} \rightarrow 2N_2O_{4(g)} + O_{2(g)}$

0.5	–	–
$0.5-2x$	$2x$	x

$P_t = P_{N_2O_5} + P_{N_2O_4} + P_{O_2}$

$= 0.5 - 2x + 2x + x$

$Pt = 0.5 + x$

$x = Pt - 0.5$

$P_{N_2O_5} = 0.5 - 2(P_t - 0.5)$

$= 0.5 - 2(0.512 - 0.5)$

$= 0.5 - 2(0.012)$

$= 0.5 - 0.024 = 0.476$

$K = \frac{2.303}{t} \log \frac{0.5}{0.476}$

$= \frac{2.303}{150} \log 1.05$

$= \frac{2.303}{150} \times 0.0212$

$= 3.25 \times 10^{-4} \text{ sec}^{-1}$.

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71. $\left[Cr(H_2O)_5Cl\right]Cl_2 + 2AgNO_3 \rightarrow 2AgCl$

$n = 0.01 \times V \qquad 0.1 \times 20$

1 mol of $\left[Cr(H_2O)_5Cl\right]Cl_2$ required 2 moles $AgNO_3$

0.1×20 mole of complex required - $0.01 \times V$ mole of $AgNO_3$

$1 \times 0.01 \times V = 2 \times 0.1 \times 20$

$V = \dfrac{2 \times 0.1 \times 20}{0.01 \times 1} = 400$ ml

72.

$$Ph-\overset{\overset{O}{\|}}{C}-CH_3 \xrightarrow[\Delta]{PCl_5} Ph-\overset{Cl}{\underset{Cl}{C}}-CH_3 + POCl_3$$

[A]

$\downarrow$ **3 eq. $NaNH_2$ / NH_3**

$$Ph-C\equiv C^{\ominus}Na^{\oplus}$$

[B]

$\downarrow$ **Acidify**

$$Ph-C\equiv C-H$$

[C]

$\downarrow$ 1. B_2H_6 2. $H_2O_2 \,|\, OH$

$$Ph-\underset{OH}{CH}=CH$$

$\rightleftharpoons$ **Tautomeriation**

$$Ph-CH_2-\overset{\overset{O}{\|}}{C}-H$$

[D]

$\Rightarrow$ Number of Sp^3C -atoms in product D = 1.

73. $Cr_2O_7^{-2}{}_{(aq)} + 14H^+{}_{(aq)} + 6e^- \rightarrow 2Cr^{+3}{}_{(aq)} + 7H_2O(l)$

$$E_{cell} = E^o{}_{cell} - \frac{0.059}{6}\log\frac{\left[Cr^{+3}\right]^2}{\left[Cr_2O_7^{-2}\right]\left[H^+\right]^{14}}$$

$$0 = 1.33 - \frac{0.059}{6}\log\frac{10^{-8}}{\left[H^+\right]^{14}} \qquad \frac{1.33 \times 6}{0.059} = \log\frac{10^{-8}}{\left[H^+\right]^{14}}$$

$135.254 = -8 - 14\log\left[H^+\right]$

$143.254 = 14P^H$

$P^H = \dfrac{143.254}{14} = 10.23.$

74. A) Rosenmund Reduction

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B) Gattermann- Koch reaction
C) friedel –crafts acylation
D) Etard reaction
E) side chain halogenation
F) preparation of carboxylic acid from grignard regents.

75. From point 1 to point 2 $(W_1 \to 2)$

Since this is an isochoric process (constant volume)

The work done, $W_1 \to 2$ is zero

$W_1 \to 2 = 0$.

From point 2 to 3 $(W_2 \to 3)$

Isobaric (constant pressure)

$W_{2\to3} = -P\Delta V$

$= -P(V_3 - V_2)$

$= -2atm \times 1Lt = -2\ L.atm$

Converting to joules (using $1Lt$ atm=101.3 J)

$W_{2\to3} = -2 \times 101.3 = -202.6$

From point 3 to point 4 $(W_{3\to4})$

Isochric process.

$W_{3\to4} = 0$

Total work = -202.6=-203.

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